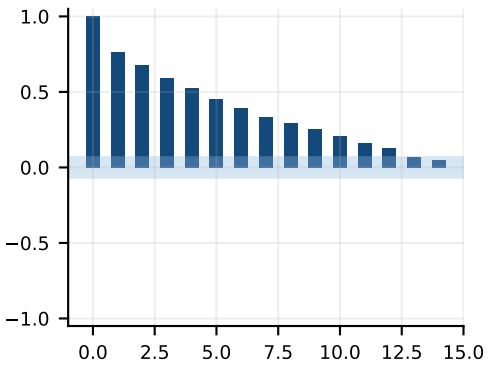


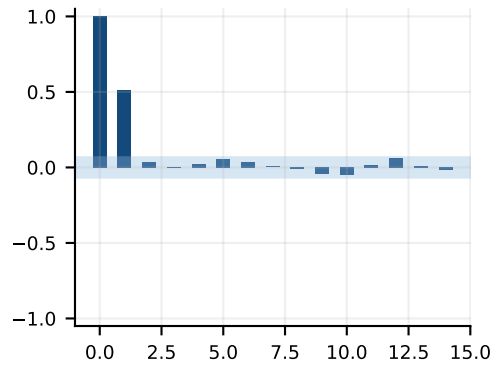
# The correlogram fingerprint

AR → PACF cuts off · MA → ACF cuts off · ARMA → both tail off · slow linear decay → difference before reading further

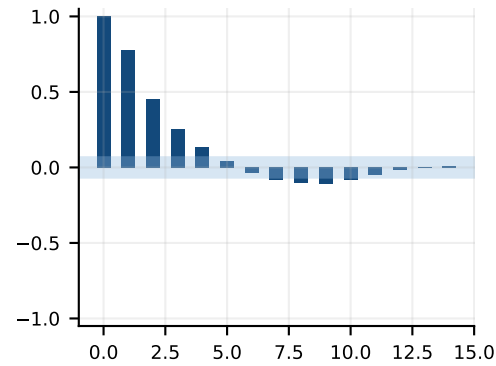
AR(2) — ACF



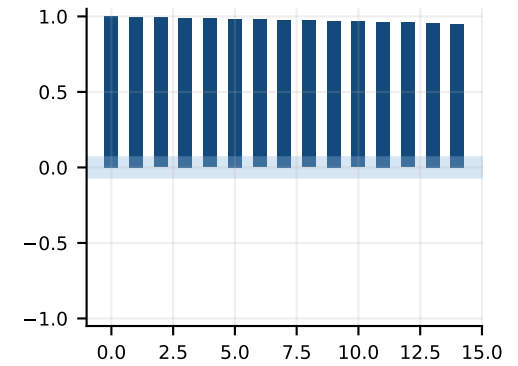
MA(1) — ACF



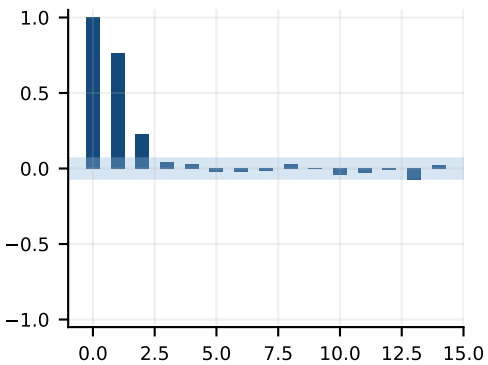
ARMA(1,1) — ACF



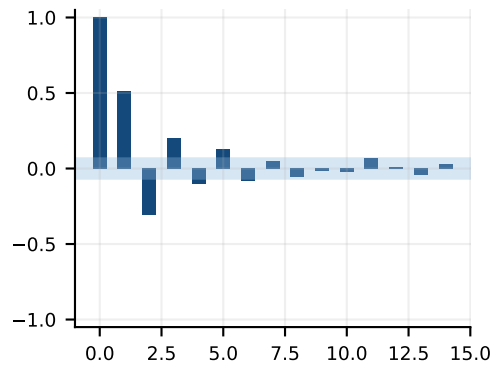
Random walk — ACF



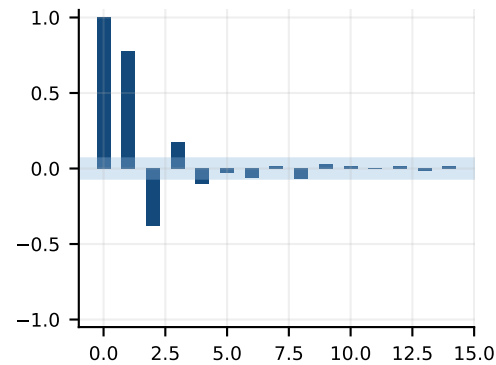
PACF



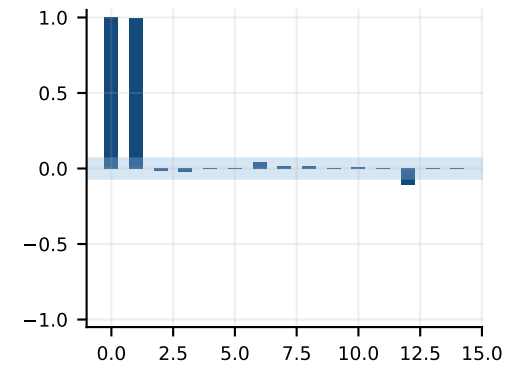
PACF



PACF



PACF



**PACF cuts off after lag 2  
→ AR(2)**

**ACF cuts off after lag 1  
→ MA(1)**

**both tail off  
→ ARMA; use AICc/BIC**

**ACF decays slowly and linearly  
→ NOT stationary. Difference first.**

*"Cuts off" means it drops inside the  $\pm 1.96/\sqrt{n}$  band. The default lags=20 never reaches lag 24 — set lags  $\geq 2m$  or the seasonal spike is invisible.*